PEDIBIRN STUDY

Abusive Head Trauma Research

Full Study Workflow, Causal Framework & All Six DAGs Explained

18 Sites · 2011–2021 · $n = 701$

The Story: Dataset & Study Design

How the PediBIRN dataset was built and what question it answers

973	701	18	38	10 yrs
Full Dataset Patients	With Retinal Exam	Academic Centres	Standardised Variables	Study Period 2011–21

Who was studied?

Children under 3 admitted to PICUs with acute symptomatic closed head injury. MVC injuries and pre-existing brain abnormalities excluded.

What is the outcome?

Severe retinal injury: haemorrhage described as dense, extensive, extending to ora serrata, and/or retinoschisis — confirmed by an ophthalmologist.

Analytic approach

Multiple mediation analysis: separates direct effects from indirect effects through mediators, across five competing causal models.

Why restrict to 701?

Only children with a documented ophthalmologist retinal exam — the outcome required expert eye assessment.

Core question

Among competing explanations — inertial injury, brain injury, hypoxia — which is the **most plausible primary cause**?

Data quality

Prospective design with monthly site audits; 38 standardised variables collected uniformly across all 18 centres.

DAGitty Model Legend — How to Read the DAGs

All six DAGs use the same visual language from the DAGitty software

Node types

Exposure
yellow-green

Outcome
dark blue

Mediator
cyan

Confounder
pink

Arrow types

Green = causal path

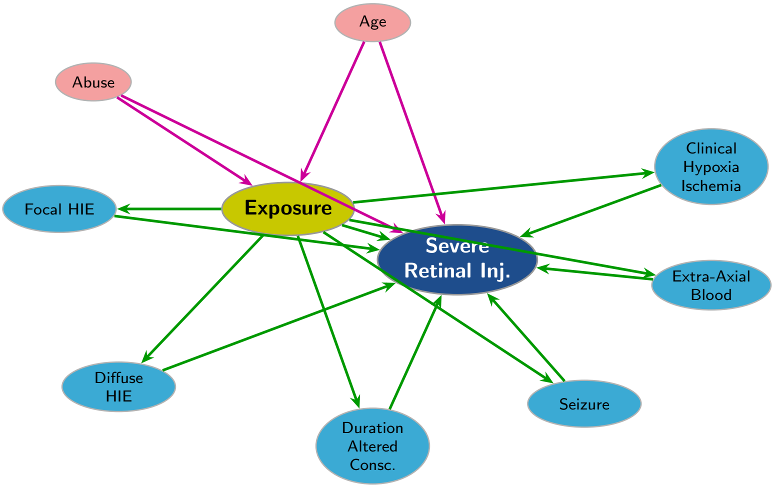
Pink = backdoor path

How to read each DAG

- The **yellow-green oval** is the **primary exposure** being tested as the causal explanation for severe retinal injury
- The **dark blue oval** is always **Severe Retinal Injury** — the fixed outcome across all models
- **Cyan ovals** are **candidate mediators** — they carry the indirect effect from exposure to outcome
- **Pink ovals** (Age, Abuse) are **confounders** — pink arrows show backdoor paths that must be blocked by conditioning

DAG 1 — General Conceptual Framework

Not a formal model — shows the mediation logic common to all five estimated models

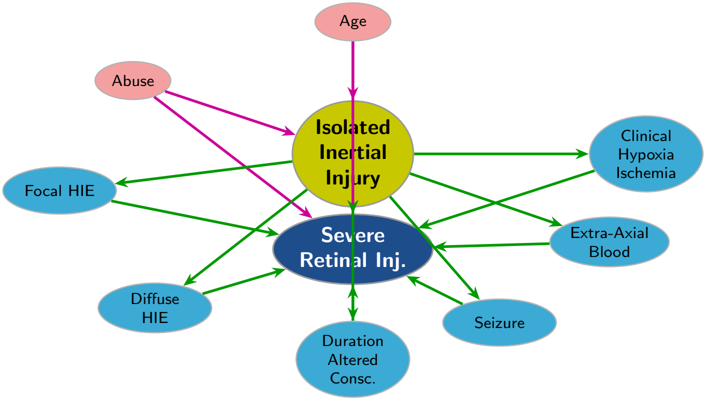


Key idea: The exposure can reach the outcome *directly* (green arrow from Exposure →

DAG 2 — Isolated Inertial Injury

★ STRONGEST MODEL

Authors' preferred model — Comparator: contact injury without inertial injury — Largest total & direct effect



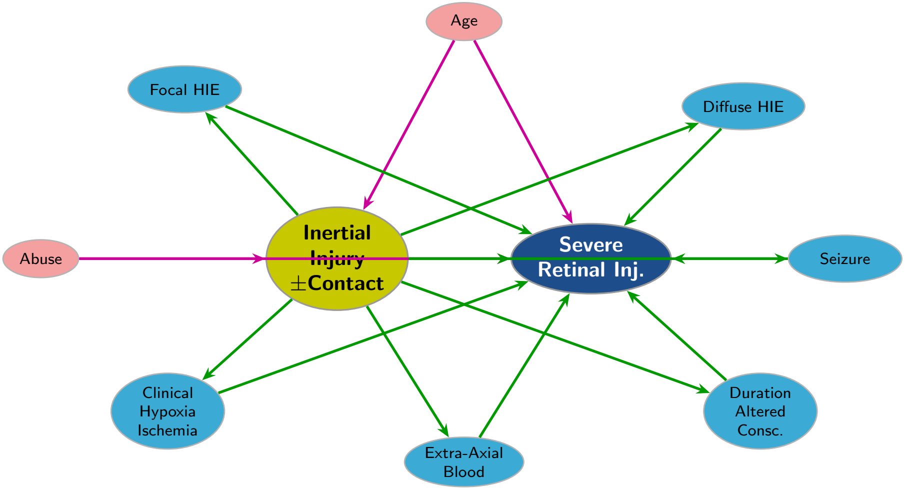
Key findings: Largest total & direct effect of all five models. Partial mediation via Diffuse HIE and Extra-Axial Blood (severity-threshold markers). All 6 mediators tested; 2 are significant.

Authors: shaking is the most parsimonious explanation.

DAG 3 — Inertial Injury (±Contact)

STRONG MODEL

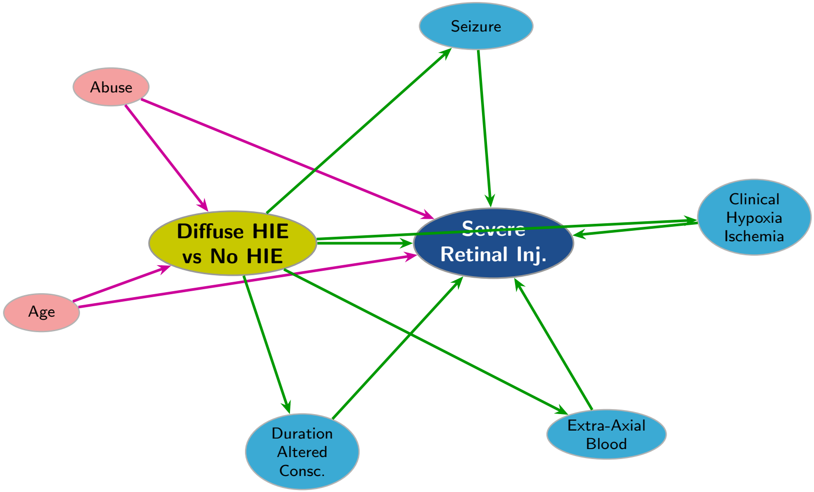
Broader inertial model — Comparator: contact injury, no inertial — 4 active mediators vs. 2 in DAG 2



DAG 4 — Diffuse HIE vs. No HIE

MODERATE MODEL

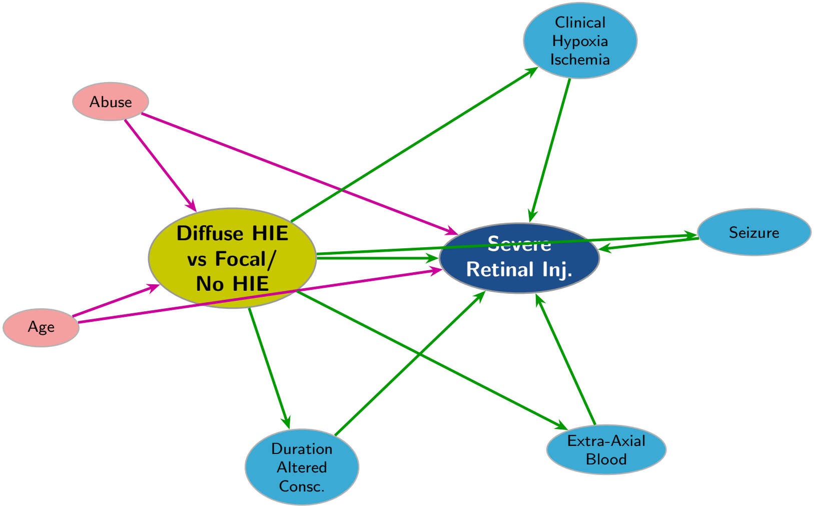
Alternative brain-injury explanation — Comparator: no HIE anywhere — >50% of effect is mediated



DAG 5 — Diffuse HIE vs. Absent or Focal HIE

MODERATE MODEL

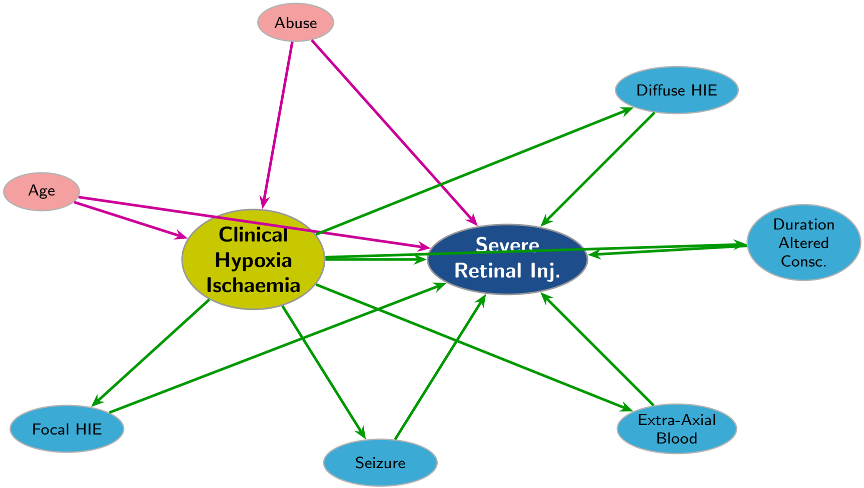
Does diffuse pattern specifically matter? — Comparator: focal HIE or no HIE — Still weaker than inertial



DAG 6 — Clinical Hypoxia/Ischaemia

× WEAKEST MODEL

Weakest explanation — Comparator: no pre-hospital hypoxia/ischaemia — ≈70% mediated



Comparative Summary — All Five Mediation Models

Ranked by explanatory strength

#	Primary Exposure	Comparator	Direct Effect	Active Mediators (n)	Best?
2	Isolated Inertial Injury	Contact, no inertial	**LARGEST**	Diffuse HIE, Extra-axial Blood (2)	✓
3	Inertial Injury (±Contact)	Contact, no inertial	Strong (< DAG 2)	Extra-axial, Hypoxia, Diffuse HIE, Seizure (4)	—
4	Diffuse HIE vs. No HIE	No HIE	<50% of effect	Altered Consc., Hypoxia, Extra-axial, Seizure (4)	—
5	Diffuse HIE vs. Focal/No HIE	Focal or No HIE	Moderate	Altered Consc., Hypoxia, Extra-axial, Seizure (4)	—
6	Clinical Hypoxia/Ischaemia	No hypoxia	**SMALLEST (≈30%)**	Diffuse HIE, Focal HIE, Altered Consc., Seizure, Extra-axial (5)	—

Key insight: DAG 2 has the largest direct pathway. DAG 6 relies almost entirely on intermediates (≈70% mediated). The same 6 nodes appear in each model — only their *role* (exposure vs. mediator) shifts between DAGs.

Conclusions & Limitations

Key Conclusions

- ★ **Isolated inertial injury** (DAG 2) is the most robust primary cause — largest total & direct effects across all models
- ◇ **Diffuse HIE & extra-axial blood** partly mediate the inertial effect — likely marking a severity threshold, not acting as independent causes
- ○ **Diffuse HIE as exposure** (DAGs 4–5) is significant but weaker — brain pathology alone does not match the explanatory power of inertial biomechanics
- × **Clinical hypoxia/ischaemia** (DAG 6) is the weakest — ≈70% mediated; a consequence rather than a primary driver

Important Limitations

- ! **Cross-sectional data** — temporal ordering cannot be observed; supports causal inference, cannot prove causation
- ! **Proxy variables only** — inertial injury, contact injury, and biomechanics are never directly measured; all inferred from clinical and radiological patterns
- ! **No direct observation of shaking** — all biomechanical inferences rely on the clinical footprint of forces
- ! **Mediation $\neq$ causation** — results should inform clinical reasoning, not legal conclusions